

Wall Climbing Inspection Robot

Abstract

This project presents a suction-based wall climbing robot developed for inspection of vertical structures such as building facades, bridges, and industrial installations. The prototype demonstrates reliable wall adhesion and controlled vertical mobility, enabling close-range inspection of hard-to-reach surfaces through remote operation.

An embedded ESP32-based vision module supports AI-assisted surface inspection, enhancing defect identification during operation. The system aims to reduce human risk in manual inspection tasks while improving accessibility and operational safety. Its modular and scalable architecture makes it suitable for further industry-driven enhancement and deployment in infrastructure monitoring applications.